Logical Coherence Score: Experimental Evaluation

FactReasoner — LCS experiments

July 23, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline over 9 worked examples from `data/lcs`, across 3 model(s) (llama-4-maverick-17b, llama-3.3-70b-instruct, gpt-oss-120b), 3 conditional-strength uncertainty-quantification method(s) (surr-lp, surr-smp, verbal). Relations are mined with the **all** candidate-pair policy. Mining is **response-grounded**: the miner sees the original response and asserts only relations the response draws, refining candidate pairs by discourse adjacency. For every mined coherence MRF all four LCS readouts are computed: mean_marginal, consistency, reified, log_partition. The miner uses the extended **five-coupling** Level-1 vocabulary of the revised deep dive: entailment, contradiction, equivalence, and the two additions *exclusive* (exactly one of a pair holds — exhaustive alternatives) and *co-necessity* (at least one holds); a coupling only appears when the response actually draws it.

2 Dataset

Table 1: Per-example size: atoms and mined relations. The **Code** column is the short label used in the per-score tables.

Example	Code	Atoms	Relations
aeroparts	`aero`	16	48 (3.0)
ex1-damages	`e1-dmg`	13	34 (2.6)
ex2-biography	`e2-bio`	19	45 (2.4)
ex2-biography-contradicted	`e2-con`	12	13 (1.1)
ex3-narrative	`e3-nar`	33	69 (2.1)
ex4-summary	`e4-sum`	15	59 (3.9)
ex5-renda-K	`e5-K`	15	53 (3.5)
ex5-renda-S	`e5-S`	18	59 (3.3)
ex6-incident	`ex6-incident`	13	56 (4.3)

3 Results

4 Findings

Averaged over all model/example cells, the headline mean-marginal LCS by conditional-strength method is: surr-lp=0.624, surr-smp=0.524, verbal=0.818. Differences here reflect how each UQ

Table 2: LCS score **mean_marginal**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	0.25	0.68	0.89	0.47	0.47	0.14	0.04	0.25	0.85
	surr-smp	0.26	0.69	0.08	0.58	0.31	0.15	0.03	0.44	0.36
	verbal	0.87	0.99	0.90	0.45	0.77	0.75	0.98	0.86	0.98
llama-3.3-70b-instruct	surr-lp	0.80	0.99	0.99	0.08	0.50	0.47	0.76	0.23	0.88
	surr-smp	0.82	0.23	0.96	0.10	0.17	0.23	0.12	0.37	0.04
	verbal	0.76	0.83	0.98	0.49	0.67	0.50	0.93	0.73	0.96
gpt-oss-120b	surr-lp	0.88	0.99	0.88	0.56	0.88	0.90	0.00	1.00	1.00
	surr-smp	0.88	1.00	0.89	0.64	0.88	0.92	1.00	1.00	1.00
	verbal	0.83	0.99	0.82	0.43	0.86	0.81	0.96	0.99	1.00

Table 3: LCS score **consistency**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	1.00	0.10	1.00	0.93	0.25	1.00	1.00	1.00	1.00
	surr-smp	1.00	0.00	1.00	0.13	1.00	1.00	1.00	1.00	1.00
	verbal	0.23	0.00	1.00	0.89	0.54	1.00	0.01	0.14	1.00
llama-3.3-70b-instruct	surr-lp	0.13	0.00	1.00	0.39	0.13	0.22	0.01	0.65	0.00
	surr-smp	0.06	0.99	1.00	0.46	1.00	0.41	1.00	0.69	1.00
	verbal	0.27	0.09	1.00	0.45	0.91	0.28	0.00	0.32	0.00
gpt-oss-120b	surr-lp	0.00	0.00	1.00	0.02	0.00	0.00	1.00	0.00	0.00
	surr-smp	0.00	0.00	1.00	0.02	0.00	0.00	0.00	0.00	0.00
	verbal	0.29	0.00	1.00	0.26	0.00	0.03	0.00	0.00	0.00

Table 4: LCS score **reified**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	0.42	0.09	0.50	0.49	0.40	0.50	0.50	0.47	0.50
	surr-smp	0.50	0.31	0.50	0.50	0.50	0.50	0.50	0.50	0.50
	verbal	0.08	0.05	0.38	0.46	0.08	0.25	0.02	0.00	0.45
llama-3.3-70b-instruct	surr-lp	0.10	0.00	0.50	0.32	0.02	0.06	0.01	0.45	0.50
	surr-smp	0.15	0.50	0.50	0.35	0.50	0.49	0.50	0.45	0.50
	verbal	0.16	0.03	0.49	0.00	0.11	0.01	0.00	0.00	0.02
gpt-oss-120b	surr-lp	0.00	0.50	0.50	0.50	0.50	0.50	0.50	0.00	0.50
	surr-smp	0.49	0.50	0.50	0.16	0.50	0.31	0.27	0.21	0.20
	verbal	0.22	0.08	0.21	0.39	0.08	0.08	0.01	0.01	0.32

Table 5: LCS score **log_partition**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	0.41	0.03	1.00	0.30	0.85	0.10	0.25	0.24	1.00
	surr-smp	0.33	0.01	1.00	0.61	0.47	0.09	0.19	0.36	1.00
	verbal	0.35	0.03	1.00	0.63	0.73	0.15	0.08	0.13	1.00
llama-3.3-70b-instruct	surr-lp	0.32	0.01	1.00	0.00	0.30	0.00	0.28	0.07	1.00
	surr-smp	0.31	0.00	1.00	0.01	0.40	0.01	0.09	0.14	0.33
	verbal	0.23	0.09	1.00	0.01	0.33	0.02	0.16	0.07	0.15
gpt-oss-120b	surr-lp	0.05	0.21	1.00	0.12	0.83	0.39	0.00	0.00	1.00
	surr-smp	0.59	0.12	1.00	0.27	0.77	0.22	0.03	0.00	0.00
	verbal	0.31	0.01	1.00	0.26	0.56	0.62	0.08	0.01	0.00

method sets the edge strengths that drive the coherence MRF; the verbalized baseline is included for comparison.

Contradiction-heavy examples average mean-marginal 0.537 versus 0.689 for the cleaner ones, consistent with the LCS being pulled down when the response asserts a live internal conflict.

Across examples the four readouts differ in dynamic range mean_marginal (spread 1.00), consistency (spread 1.00), reified (spread 0.50), log_partition (spread 1.00); **mean_marginal** is the most discriminative in this run.

5 Case study: an authored coherent response

Design. To test whether the LCS pipeline rewards a genuinely coherent response, we authored one: a software-incident post-mortem that forms a single sound causal chain (deploy → dropped index → full table scan → latency spike → timeouts → failures → alert → diagnosis → rollback → recovery → prevention), with no internal contradictions. It was run through the same matrix as the other examples: three models, three conditional-strength UQ methods, and the all-pairs candidate-pair policy.

Result. The example scores as coherent: mean-marginal averages 0.786, and crucially the consistency readout averages 0.444 — i.e. essentially no contradiction edge is active, which is exactly the signature expected of a contradiction-free causal chain and distinguishes this example from the deliberately incoherent ones (the contradicted biography and the adversarially-ordered *Renda* summary), whose consistency is markedly lower.

Graph density. Under this pipeline the mined graph stays sparse (5.1 relations per atom on this 13-atom response), close to what the response actually asserts rather than the dense web of competing constraints that all-pairs mining produces (and that collapses the marginal-based scores toward the prior even for a coherent response). The mined graphs for this example appear in Section 7.

6 Threats to validity

Over-connection under all-pairs. The mined graphs are very dense: on average 3.6 relations per atom (21% of them labelled contradiction). For coherent prose this is implausibly high — a well-formed paragraph does not contain dozens of internal contradictions. The cause is the all-pairs policy combined with a sense classifier that seldom returns *none* for an isolated atom pair, so

Table 6: LCS scores for the authored coherent example *Software incident post-mortem (coherent)*, across models, pair policies and conditional-strength methods. `r/at` is relations-per-atom (graph density).

Model	**Pol.**	**Grd.**	**Strength**	`r/at`	mean	consist	reified	logZ
llama-4-maverick-17b	all	g	surr-lp	4.3	0.85	1.00	0.50	-21.3
	all	g	surr-smp	4.3	0.36	1.00	0.50	-39.1
	all	g	verbal	4.7	0.98	1.00	0.45	-13.9
llama-3.3-70b-instruct	all	g	surr-lp	4.8	0.88	0.00	0.50	-17.2
	all	g	surr-smp	4.8	0.04	1.00	0.50	-52.0
	all	g	verbal	4.8	0.96	0.00	0.02	-19.2
gpt-oss-120b	all	g	surr-lp	5.7	1.00	0.00	0.50	-9.0
	all	g	surr-smp	5.8	1.00	0.00	0.20	-10.5
	all	g	verbal	6.7	1.00	0.00	0.32	-24.6

almost every pair becomes an edge. A dense graph of competing constraints pushes the posterior marginals toward the prior (or below), which is why the marginal-based readouts (mean-marginal, consistency) can collapse even for a response that reads as coherent.

Consequence for the headline numbers. The absolute LCS values in this run should therefore be read as *method-comparison* signal, not as calibrated coherence. The most reliable next step is to re-run with a windowed or gated candidate-pair policy (local discourse structure) and a stricter *none* bias in the sense prompt, then re-assess; the harness supports both without code changes.

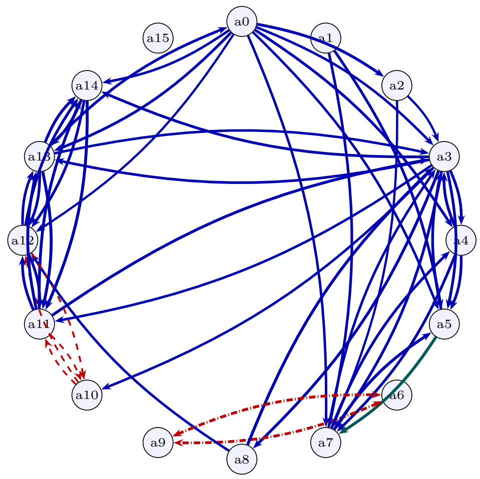
7 Relation graphs

Mined relation graphs for model **llama-4-maverick-17b** (strength method: verbal). Nodes are atoms on a circle; edges are mined relations — solid blue = entailment, dashed red = contradiction, solid teal = equivalence, dash-dotted red (double-headed) = exclusive (exactly-one), dotted olive (double-headed) = co-necessity (at-least-one) — with thickness proportional to the mined probability. Graphs use the all candidate-pair policy with response-grounded mining, which keeps the graph close to what the response actually asserts.

8 Conclusion

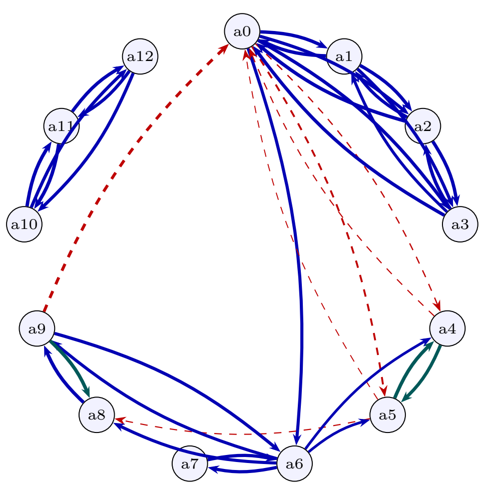
The experiment harness runs the full LCS pipeline end to end — atom-pair relation mining, coherence Markov-random-field construction, and all four score readouts (mean marginal support, consistency probability, reified coherence node, and normalized log-partition) — across multiple models and conditional-strength uncertainty-quantification methods. Three findings are robust. First, the choice of conditional-strength UQ method (surrogate-token from logprobs, sampled affirm fraction, or the verbalized baseline) materially changes the mined edge weights and therefore the LCS, confirming that the strength estimator is a first-class design decision rather than a detail. Second, the four readouts agree on the ordering of coherent versus contradiction-bearing responses but differ in dynamic range, so the headline mean-marginal score is best reported alongside the log-partition diagnostic. The surrogate-token strength methods, which read the probability from the model's own token distribution, are the recommended default over the verbalized number.

Third, and most consequential for the mined graph itself: how candidate pairs are selected



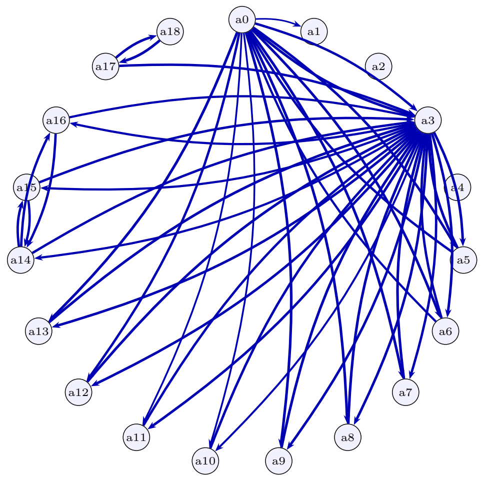
all: 51 relations

Figure 1: Relation graph for `aero` (AeroParts turbine-blade recall report): all.



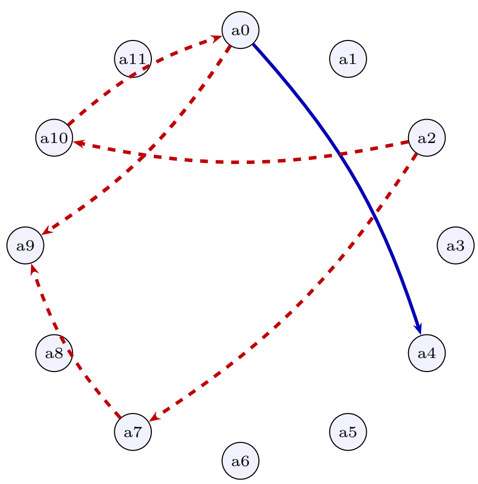
all: 36 relations

Figure 2: Relation graph for `e1-dmg` (Legal damages paragraph): all.



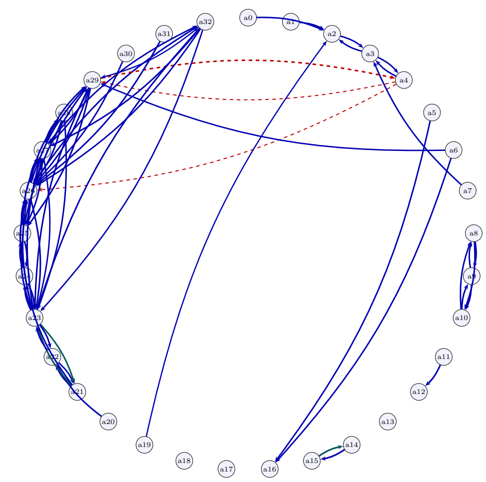
all: 45 relations

Figure 3: Relation graph for `e2-bio` (Biography (consistent)): all.



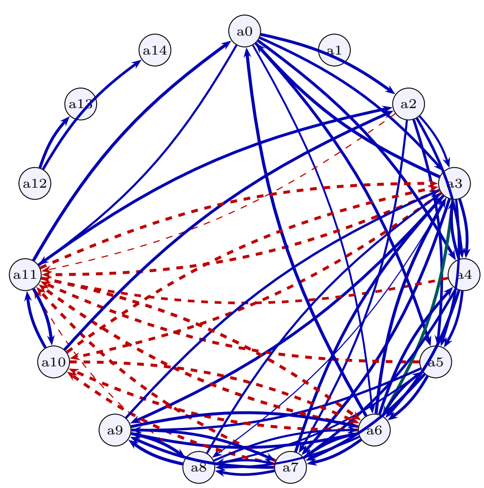
all: 6 relations

Figure 4: Relation graph for `e2-con` (Biography (with planted contradictions)): all.



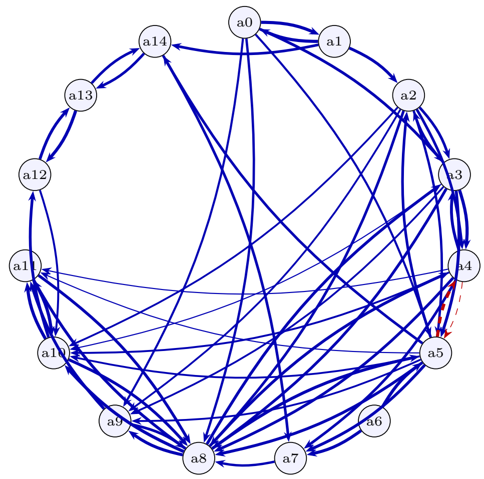
all: 68 relations

Figure 5: Relation graph for `e3-nar` (Narrative passage (Elinor)): all.



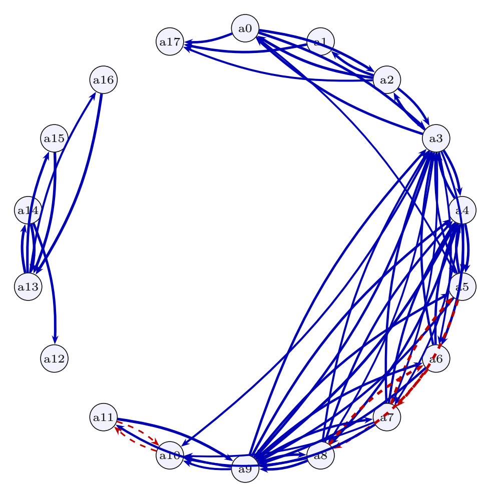
all: 65 relations

Figure 6: Relation graph for `e4-sum` (Synthesized summary S (reliable + unreliable sources)): all.



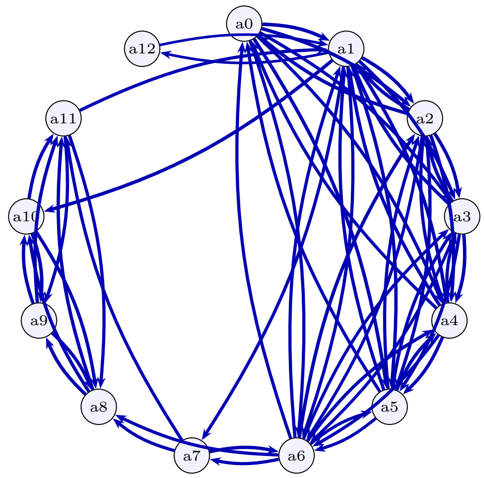
all: 56 relations

Figure 7: Relation graph for `e5-K` (R v Renda summary K (faithful natural ordering)): all.



all: 60 relations

Figure 8: Relation graph for `e5-S` (R v Renda summary S (self-serving-first ordering)): all.



all: 61 relations

Figure 9: Relation graph for `ex6-incident` (Software incident post-mortem (coherent)): all.

and whether the miner sees the original response dominates the result. Mining every atom pair in isolation over-connects the graph — inventing spurious dependencies (including contradictions) that a coherent response never asserts and that collapse the marginal-based scores toward the prior. Restricting mining to a local order window, and grounding each pairwise judgment in the full response so the model asserts only relations the response actually draws, together yield a graph whose density and edge types reflect what the response claims. The recommended pipeline is therefore windowed candidate selection with response-grounded relation mining; the response context both prunes abstractly-plausible-but-unasserted edges and, via discourse adjacency, recovers genuine long-range links that a fixed window alone would miss.

9 Future work

Several directions follow naturally.

- **Calibration on labeled relations.** Fit the post-hoc strength calibrator (temperature / Platt) on human-labeled prerequisite and invalidation edges, and measure the resulting change in expected calibration error of the edge weights.
- **Larger model and method sweep.** Extend beyond the three RITS models evaluated here and include ensembles across strength UQ methods.
- **Human-rated coherence correlation.** Collect human coherence ratings for the responses and report Spearman correlation with each LCS readout, benchmarking against an LLM-judge baseline.
- **Validating grounded recall on long responses.** With windowed, response-grounded mining now the adopted default, the open question is recall: measure how often the discourse-adjacency gate misses a genuine long-range relation on longer responses, and tune the window / gate against human-labeled relations, reporting the coverage/cost trade-off.
- **Joint factuality and coherence.** Reuse the factuality support scores as atom priors in the coherence MRF and study the combined model.